

LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.7.1 – Internet Service Types

Lesson Overview:

In this lesson, students will explore common internet service types and compare the types based on their features.

Students will:

- Identify and describe different internet service types.
- Compare internet service types based on their pros and cons.
- Analyze situations and determine which internet service type best meets a user's needs.

Guiding Question: What are the different internet service types, and what should a user consider when choosing a service?

Suggested Grade Levels: 8-12

Technology Needed: No

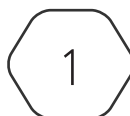
Approximate Time Required: 65-85 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.7 – Compare and contrast common internet service types.
 - Fiber optic
 - Cable
 - Digital subscriber line (DSL)
 - Wireless
 - RF
 - Satellite
 - Cellular

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Lesson Background

Background Information:

In this lesson, students are introduced to the most common types of internet services available which include fiber optic, cable, DSL, RF, Satellite, and cellular. Students should be able to identify, describe, and compare the internet types based on key factors such as the equipment needed, speed, latency, and pros and cons. By analyzing real-world scenarios, students will put their knowledge into practice by determining the best internet solution based on a user's needs and environment.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.7.1 – Internet Service Types • Activity Slides 2.7.1 – Internet Service Recommendation
Materials per Student	• Student Notes 2.7.1 – Internet Service Types • Student Handout 2.7.1 – Internet Service Types • Assessment 2.7.1 – Internet Service Types (optional)

Cyber Terms:

Term	Definition
Broadband Internet	high-speed internet connections that allow multiple devices to stay connected and data to flow quickly and smoothly
Fiber Optic Internet	a service that transmits data as light signals, allowing for fast speeds, high bandwidth, and low latency
Cable Internet	a service that uses coaxial cables to transmit data through electrical signals
Digital Subscriber Line (DSL) Internet	a service that uses telephone lines to transmit data through electrical signals
Radio Frequency (RF) Internet	a service that uses radio waves to transmit data between a base station and a receiver
Satellite Internet	a service that transmits data through satellites in space, which communicates with satellite dishes on earth that are connected to the internet
Cellular Internet	a service that uses mobile network towers to transmit data to and from mobile devices and modems

2.7.1 – Internet Service Types

Guiding Question: What are the different internet service types, and what should a user consider when choosing a service?

Lesson Launch (5 minutes)

- Introduce students to **Broadband Internet** and how all services are not the same.
- Discuss the five consideration factors when choosing a service type.
 - Availability is whether the service can be installed or used at a specific address.
 - Coverage is where a service can reach or broadcast a signal.
- **Say:** Today we will learn about the different internet service types, compare them, and decide when they best meet a user's needs.

Broadband Breakdown: Pick Your Connection (30-40 minutes)

- Distribute the Student Handout 2.7.1 – Internet Service Types and have students fill in the answers during the lesson.
- Introduce **Fiber Optic Internet**, its definition, its features, pros and cons.
 - Fiber Optic Internet offers symmetrical speeds, meaning the upload speeds can match the download speeds. This is beneficial for activities that involve significant data uploads, like video conferencing and cloud storage.
 - Fiber-to-the-Home (FTTH) runs fiber cables directly from the internet service provider's (ISP) data center to the user's residence or business.
 - Fiber-to-the-Curb (FTTC) runs fiber cables to a cabinet, placed by the ISP, on a street close by; and from the cabinet, very high bit-rate digital subscriber line (VDSL) cables are used to connect to the user's residence or business.
 - Pros:
 - Fiber Optic Internet offers some of the fastest speeds available, it is ideal for high-demand applications and streaming.
 - Fiber Optic Internet is reliable over long distances and is resistant to interference from electrical signals or weather conditions.
 - Cons:
 - Fiber Optic Internet is not universally available, deployment is concentrated in urban and suburban areas, leaving rural and similar areas without the option.
 - The installation and setup costs of Fiber Optic Internet are relatively higher compared to other types of internet services. Monthly service fees can also be more expensive but vary based on the ISP and region.

- Introduce **Cable Internet**, its definition, its features, pros and cons.
 - Cable Internet offers asymmetrical speeds where download speed is faster than upload speed.
 - The cables connect to a cable modem to convert RF signals into digital data for a router.
 - Cable Internet provides broadband access with the Data Over Cable Service Internet Specification (DOCSIS) standard, used for data transmission.
 - The use of coaxial cables started with cable TV, but now it can also be used for Cable Internet.
 - Pros:
 - Cable Internet is widely available in many urban and suburban areas; it can cover a broader region due to existing cable TV infrastructure.
 - Installation is straightforward, especially for users that already have cable TV service.
 - Cons:
 - Cable Internet bandwidth is shared among users in a particular neighborhood; during peak usage times, speeds can decrease due to network congestion.
 - Cable Internet can experience fluctuations in performance due to shared bandwidth and interference from electrical devices.
- Introduce **Digital Subscriber Line (DSL) Internet**, its definition, its features, pros and cons.
 - DSL Internet separates voice and data signals by using different frequency bands, allowing simultaneous use of voice calls and internet access through a phone line.
 - There are several types of DSL including:
 - Asymmetric DSL (ADSL): It provides faster download speeds than upload speeds and is common for residential use.
 - Symmetric DSL (SDSL): It offers equal download and upload speeds; it is useful for businesses that require balanced data performance.
 - Very high bit-rate DSL (VDSL): It provides higher speeds than ADSL but over shorter distances; it is used for demanding applications.
 - Pros:
 - DSL Internet can be provided over existing telephone lines, making it available in many areas including rural and remote locations where fiber optic or cable internet may not reach.
 - DSL Internet is typically less expensive than cable and fiber optic options.
 - Cons:
 - DSL Internet speeds are lower compared to fiber optic and cable.
 - The quality and speed of DSL Internet decreases with its distance from the telephone company's exchange or central office.
- Introduce **Radio Frequency (RF) Internet**, its definition, its features, pros and cons.
 - Receivers are wireless routers or antennas.

- This method involves a fixed wireless setup where the receiver is installed at a specific location, such as a home or business.
- RF uses a special outdoor antenna that can handle all kinds of weather from big storms to heavy snow.
- Base stations, often mounted on towers or tall buildings, broadcast signals to multiple receivers in a coverage area.
- The internet signals are sent from towers using RF waves through the air straight to where you live or work.
- The antenna must be within 10 miles of the tower without anything blocking their line-of-sight.
- RF Internet is also known as fixed wireless internet.
- Pros:
 - RF Internet can provide internet access in areas where wired infrastructure is not available, such as rural or remote locations.
 - RF Internet is generally faster to deploy than laying cables, especially in areas with difficult terrain.
- Cons:
 - RF Internet requires a clear line-of-sight between the base station and the receiver, buildings or trees can interfere with the signal.
 - Speeds can vary based on the distance from the base station and environmental conditions.
- Introduce **Satellite Internet**, its definition, its features, pros and cons.
 - Satellite Internet is mostly found in areas where traditional internet doesn't work well or isn't available.
 - Satellite does not rely on cables or other physical connections, other than the modem and router.
 - The satellite dish must be aimed precisely at the orbiting satellite for a strong signal.
 - The internet signals must travel a long way due to the distance between the receiving dish and the satellite in space.
 - Pros:
 - Satellite Internet is useful in areas where it is difficult or impossible to run cable, including remote and rural areas.
 - Satellite Internet will allow one orbiting satellite to provide a signal to several receivers simultaneously.
 - Cons:
 - Satellite Internet has a high latency and low speed due to the long distance that data must travel to and from the satellite.
 - Weather conditions can impact the signal quality and reliability of Satellite Internet.
- Introduce **Cellular Internet**, its definition, its features, pros and cons.

- The data is transmitted over cellular networks like 4G, LTE, and 5G.
- Mobile devices have a built-in modem. Routers are not needed with the cell device. If using cellular internet at home with a non-mobile device, a router and modem is necessary.
- Devices that use Cellular Internet receive radio waves to connect to both the internet and telephone network.
- Cell towers can provide coverage from a few hundred feet up to five miles and can support high-speed data transmission.
- Obstacles like trees or buildings can block or weaken signals.
- Pros:
 - Cellular Internet allows internet access on-the-go with mobile devices or portable hotspots.
 - Cellular Internet is available in most urban and suburban areas, and increasingly in rural areas as 5G networks expand.
- Cons:
 - Cellular Internet often comes with data limits after a certain amount of usage.
 - Cellular Internet performance can be affected by physical obstructions, such as buildings or geographical features, which can impact signal strength.

Activity: Internet Service Recommendation (20-25 minutes)

- Display Activity Slides 2.7.1 – Internet Service Recommendation, have students record their answers in the handout.
- Choose if you want students to work together or in a small group.
- Read each scenario aloud or let students read independently or in their group.
- Students will analyze the scenario and determine which internet service best meets a user's needs, the equipment required, and the trade-offs of choosing that service.
- Students should consider the user's location, speed requirements, equipment availability, and budget.
- Provide students with time to write their answers.
- It is ok if student answers do not match suggested answers, if the answers are justified and reasonable.

Scenarios and Answers

- Jennifer lives in a suburban neighborhood where cable and fiber optic are available. She works from home and often joins video conferences, uploads large files, and streams HD videos. She wants a reliable, fast connection with low latency. Which internet service would you recommend and why? What equipment does the user need and what are the trade-offs for choosing that service?

Jennifer should get fiber optic internet because it offers the highest speeds and lowest latency. It is perfect for video conferencing, large file uploads, and streaming. She would need fiber optic cables and a router; the trade-offs are it being more expensive than other options.

- Liam lives in a rural valley where there are no cable or fiber lines. However, there is a wireless tower on a hill nearby and his house is in clear line-of-sight. He needs the internet for remote work, video calls, and streaming movies. Which internet service would you recommend and why? What equipment does the user need and are the trade-offs for choosing that service?

Liam should use RF internet because his house is in the line-of-sight of a wireless tower, he is also close to the tower which will allow him to get decent speeds. He will need an outdoor antenna, a modem, and router; tradeoffs are that weather could impact the signal.

- Maria lives in an apartment building downtown. She wants an affordable internet plan mostly for streaming movies and casual web surfing. She has access to cable and DSL services. Which internet service would you recommend and why? What equipment does the user need and what are the trade-offs for choosing that service?

Maria should use cable internet because it offers faster speeds than DSL and is usually affordable in urban areas. She would need a coaxial cable, modem, and router; trade-offs are that she will have a shared bandwidth which can reduce her speeds during peak times.

- Zeke lives in a rural area with large mountains and no wired internet options. He needs internet for online classes and video calls but experiences frequent weather issues like heavy rain and snow. Which internet service would you recommend and why? What equipment does the user need and what are the trade-offs for choosing that service?

Zeke should use satellite internet because it is the best option for remote areas that cannot run wires, and it does not need line-of-site with a tower to function properly. He would need a satellite dish, modem, and router; trade-offs are that he would have high latency with low speeds, and his connection could be affected by weather.

- Ellie lives in a small town with the ability to run physical cables. She mostly checks social media, streams music, and works on documents online. She doesn't need very high speeds but wants a stable connection and is on a budget. Which internet service would you recommend and why? What equipment does the user need and what are the trade-offs for choosing that service?

Ellie should use DSL internet because it is widely available due to existing phone lines, it is sufficient for her basic needs, and cost-effective. She would need telephone wires, a modem, and router; trade-offs are slower speeds and higher latency compared to other options, and distance from the ISP can affect connection performance.

- Nina travels frequently for work and often stays in hotels or remote locations where wired internet isn't available. She needs fast, reliable service for email, video conferencing, and occasional file downloads. Which internet service would you recommend and why? What equipment does the user need and what are the trade-offs for choosing that service?

Nina should use cellular internet because it is wireless and will allow her to access internet on-the-go through her mobile device. She will need her mobile device and cell towers since she is using cellular internet via her phone; trade-offs are data limits after a certain amount of usage and signal interference from weather or buildings.

Putting It All Together (10-15 minutes)

- Review what students learned today by discussing the questions below as a class.
- What five factors should a user consider before choosing an internet service type?

A user should consider equipment, speed, latency, availability and coverage, and costs and data limits before choosing an internet service.

- What are fiber optic cables and how do they transmit data?

Fiber optic cables have a central core of clear plastic or glass with a layer of reflective cladding that acts like a mirror. These cables transmit data as light signals.

- What are the main differences between Cable and DSL internet?

Cable internet uses cable TV connections to connect to homes or businesses while DSL internet uses telephone lines to connect to homes or businesses.

- What are the main differences between Radio Frequency and Satellite internet?

RF internet uses antennas and radio waves to transmit data to homes or businesses while Satellite internet uses satellites in space and satellite dishes on Earth to transmit data between homes or businesses.

- How does Cellular Internet work?

Cellular Internet uses mobile network towers to transmit data to and from mobile devices or modems.